Commitment and present bias: Experimental evidence*

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Working paper. Latest version available here.

Abstract People routinely plan to act patiently and then favor the immediate reward when the moment arrives. In a laboratory experiment, participants make a sequence of intertemporal allocation decisions under one of two conditions. We estimate a quasi-hyperbolic discounting model and find that the treatment lowers the present-bias parameter and raises demand for a commitment device. The results suggest that small changes in choice architecture can meaningfully shift self-control.

***JEL* Codes** C91, D15, D91

Keywords present bias, commitment, intertemporal choice, laboratory experiment

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* For helpful comments I thank seminar audiences and colleagues; Roe and Smith (2019) generously shared data and code. This study was approved under IRB protocol 0000 and pre-registered at the AEA RCT Registry (AEARCTR-0000000).

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1 Introduction

People often plan to act patiently and then choose the immediate reward when the moment arrives. We ask whether a simple change in how future rewards are presented amplifies this present bias, and whether it raises the demand for commitment devices.

Cite in parentheses with (Doe 2020), or in text with Lee, Kim, and Park (2021). Multiple sources collapse neatly: (Doe 2020; Roe and Smith 2019; Lee, Kim, and Park 2021). For an author whose name is already in the sentence, print only the linked year, as Doe (2020) shows. Cross-reference sections, equations, tables, and figures with `cleveref`: see section 2, equation (1), table 1, and figure 1.

2 Model

A decision maker values a stream of consumption using quasi-hyperbolic discount factors. The indicator macro `\mybb{1}(...)` typesets a blackboard "1", e.g. $\mathbb{1}(x > 0)$. The objective is a displayed, labelled equation:

$$U_t = u(x_t) + \beta \sum_{\tau=1} \delta^\tau u(x_{t+\tau}). \tag{1}$$

A second equation referenced together with the first (equations (1) and (2)):

$$\frac{\partial U_t}{\partial x_t} = 0. \tag{2}$$

3 Hypotheses

State each pre-registered prediction in the `hypothesis` environment so it is numbered and `cleveref`-referable (see Hypothesis 1):

Table 1: Example results table

	Model 1	Model 2
Treatment effect	0.281	0.895
standard error	0.044	0.052
Observations	897	897

Note: Estimated treatment effects, with standard errors in the row below. Numbers in the `d{1.3}` columns align on the decimal point; wrap headers and integer cells in `\multicolumn{1}{c}{...}`.

Hypothesis 1 *The treatment lowers the present-bias parameter relative to the control:* $\beta_{treat} < \beta_{control}$.

Hypothesis 2 *Participants in the treatment are more likely to take up the commitment device.*

4 Methodology

A custom-labelled enumeration (e.g. for a multi-step estimation procedure):

Step 1: Estimate β and δ for each participant by maximum likelihood from the allocation choices.

Step 2: Compare the estimated β across conditions, reporting Roe–Smith (2019) standard errors to account for the two-stage estimator.

5 Results

Table 1 and figure 1 report the main results.

(a) First panel

(b) Second panel

Figure 1: Example figure with two panels

6 Conclusion

A simple change in how future rewards are presented shifts the estimated present-bias parameter and raises commitment demand. We note the usual caveats of a single laboratory sample and point to field replication as the natural next step.

References

Doe, Jane. 2020. “A representative single-author article.” *Journal of Placeholder Studies* 42 (3): 101–128.

Lee, Alex, Bo Kim, and Chris Park. 2021. “A three-author article.” *Quarterly Journal of Templates* 15 (2): 55–92.

Roe, Richard, and Susan Smith. 2019. “A two-author article, for the en-dash citation example.” *Review of Example Economics* 7 (1): 1–40.

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Supplement for online publication

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Contents

S.1 First supplement section

Supplement sections are numbered S.1, S.2, … and appear in the contents above. Reference them like supplement section 1 or supplement section 2.

S.1.1 A subsection

Numbered S.1.1.

S.1.1.1 A sub-subsection

Numbered S.1.1.1 (not added to the contents by default).

S.2 Additional results

Supplement floats are numbered with an "S." prefix automatically: see table S.2 and figure S.2.

Table S.2: A supplementary table

Row	Column A	Column B
First	1	2
Second	3	4

Note: An example supplementary table.

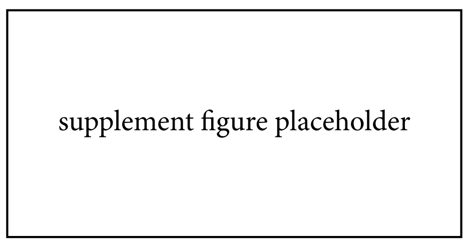

Figure S.2: A supplementary figure